

Project Title: Movie Recommendation Expert System
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1. Introduction

This project is a Movie Recommendation Expert System created using Prolog programming language to help the user in picking out a movie based on user preferences, current mood, and viewing history. Most users find it hard to choose a movie and often take time, that's why this expert system is important. It doesn't not only help you choose a movie based on preferences but also shows the location of the cinema and the availability of the movie and ensures that the movie suggestions are highly relevant to the user's specific context, and their proximity to a local cinema.

2. Objectives

- **Logic Integration:** To implement an inference engine using Prolog rules (like Cut and Negation) for accurate decision-making.
- **Personalization:** To deliver movies suggestions by filtering out user history and matching current mood/genre preferences.
- **Logistics Mapping:** To bridge the gap between movie selection and viewing by identifying specific local cinema locations.
- **System Continuity:** To develop a recursive user interface that handles multiple queries and invalid inputs without restarting.

3. System Description & Implementation

Our system idea is an expert system developed using **SWI-Prolog** that recommends movies to users based on their preferences. We gathered facts about movies such as genre, duration, mood, rating, and cinema location. The system asks the user questions and efficiently searches for the most suitable movie according to their answers. It is designed for cinema customers of all ages who want quick and personalized movie recommendations. The main tool used to build the system is **SWI-Prolog**.

- Prolog's **backtracking mechanism** allows the system to search through the knowledge base (movie/5 facts) efficiently until it finds a movie that matches all user constraints.
- The constraints checked simultaneously include:
 - Genre
 - Duration
 - Mood
 - Logical negation of the watched list (movies already seen are excluded).

- The **cut operator (!)** in the `find_movie` rule is an important feature because it stops further searching after the first suitable movie is found.
- This ensures the system gives **only one recommendation**, creating a clearer and more concise user experience.
- The **recursive loop** in `run_main_menu` creates a state-like behavior, even though Prolog is naturally stateless.
- By passing the Movies list in each iteration, the system can **remember previously watched movies** during the session.

4. Brief explanation of the code

Knowledge Base (Facts):

- **Movie Database:** Contains a diverse list of movies categorized by Genre (Sci-fi, Comedy, Animation, Horror, Action, Romance), Duration (Short/Long), Mood (Thoughtful, Cheerful, Scared, Excited), and Rating.
- **Cinema Mapping:** A set of facts linking movies to specific local cinemas such as Empire Cinemas, AMC Cinemas, and MUVI Cinemas.

Key Features & Programming Logic:

- **User History Tracking:** Uses a recursive rule (`watched/1`) to collect a list of movies the user has already seen, ensuring the system doesn't recommend something they've already watched.
- **Personalized Recommendation Engine:** Implements the `find_movie/`
- **Quality control feature:** Provides a high-standard filtering system by prioritizing top-rated movies with **4.5 stars or higher**.

5. Testing result

The image displays two screenshots of the SWISH programming environment, showing the execution of a movie recommendation program.

Top Screenshot:

- Program Editor:** Shows the source code for a movie recommendation system. The code includes a menu controller, a function to display a personal recommendation, and a function to find movies based on user input.
- Execution Window:** Displays the program's output. It shows the main menu, the user's selection of a personal recommendation, and the program's response, which includes a list of recommended movies and a prompt for the user to enter a movie name.

Bottom Screenshot:

- Program Editor:** Shows the same source code as the top screenshot.
- Execution Window:** Displays the program's output, showing a list of recommended movies and instructions for the user. The user has entered a movie name, and the program has responded with a list of movies and a prompt for the user to enter a movie name.

SWISH

File

Edit

Examples

Help

286 words written

Search

Program

171

;

watched(Rest),

172

Movies = [Film | Rest]

173

).

174

).

175

/*

176

execute_option/2: Menu controller.

177

Facilitates dialogue between the user and the system using read() and write() [c

178

*/

179

execute_option(1, Movies) :-

180

nl, write('--- Personal Recommendation ---'), nl,

181

write('Genre (sci-fi, comedy, animation, horror, action, romance): '), read(G)

182

write('Duration (short/long): '), read(L),

183

write('Mood (thoughtful, cheerful, scared, excited): '), read(M),

184

find_movie(G, L, M, Movies).

185

execute_option(2, Movies) :-

186

nl, write('--- Top Rated (4.5+) ---'), nl,

187

(best_rating(Title, Rating, Genre, Movies),

188

write(' - '), write(Title), write(' ['), write(Genre), write('] - Rating: '

189

fail % Forces backtracking to find ALL top rated movies

190

; true

191

).

192

).

193

execute_option(3, _) :-

194

nl, write('Which movie are you looking for? '), read(Title1),

195

(in_cinema(Title1, Cinema) ->

196

nl, write(Title1), write(' is playing at: '), write(Cinema), nl

197

; nl, write('Sorry, that movie is not currently in our local cinema database

198

).

199

).

200

% Handles input errors [cite: 1]

201

%

202

%

Mood (thoughtful, cheerful, scared, excited):

excited

Based on your mood and preferences, we recommend: The Dictator

Main Menu:

1. Get a personalized movie recommendation

2. View top-rated movies (4.5+)

3. Find where a movie is currently airing

4. Exit the system

Selection:

3

Which movie are you looking for?

The Dictator

The Dictator is playing at: AMC Cinemas

Main Menu:

1. Get a personalized movie recommendation

2. View top-rated movies (4.5+)

3. Find where a movie is currently airing

4. Exit the system

Selection:

4

Thank you for using the system. Goodbye!

true

Next 10 100 1,000 Stop

start.

Examples History Solutions

Table results Run